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Daniel Liu

About Me

Professional Interests: High Performance Computing, Data Science, System Design & Scalability, Computational Nuclear Physics
Languages: Python, C (GCC/clang), C++, Matlab, JavaScript, SQL
Technologies: Git, Unix, Vi, AWS/GCP, Jupyter/pandas, matplotlib, XGBoost, LightGBM, MCNP & Serpent
Legal Status: Canadian Citizen. Available on TN/H1B visa. Green Card in progress.

Experience

Terrestrial Energy
Toronto
Technical Lead, Radiation Protection & Physics
January 2020 – Present

- Developing engineering parameters for a 450 MW_{th} molten salt reactor using monte carlo physics codes in Python & C. Executing simulations to develop a full-scope simulator of the reactor by establishing nuclear physics parameters
- Owner of neutronics code (EQL0D & Serpent) used to simulate the evolution of fuel burnup in Molten Salt Reactors for fuel cycle modeling & long-term waste disposal
- Designing shielding geometric dimensions & tolerance requirements at various receptor point locations using SCALE, MicroShield, Serpent

Ontario Power Generation
Toronto
Project Engineer, D2O Storage Project
September 2018 – December 2019

- Lead infrastructure commissioning for a $500M+ nuclear project used to store radioactive heavy water used in Canadian nuclear reactors by coordinating and executing Engineering, Procurement and Construction milestones with vendors.
- Created workplans to commission various structures, systems, and components enveloping software, computer, electrical, civil, chemical, and mechanical codes & standards to bring into service a new nuclear class building (e.g., CSA N Series, NBCC, NFCC, NFPA, IEEE, ASME BPVC/B51)

Canadian Nuclear Laboratories
Chalk River
Research Engineer, Hydrogen Isotopes R&D
January 2018 – September 2018

- Engineered a system used to enrich Tritium gas (>90%) using the Thermal Cycling Absorption Process. Automated the system using B&R based PLC's, Arduinos, and relays to automatically control process instrumentation. Designed the connection of systems using Swagelok and ConFlat connections. Researched the effectiveness of various catalysts on unique mediums (e.g., Pd/graphene, Pd/carbon nanotubes, Pd/kieselguhr, etc.)

Ontario Power Generation
Toronto
Technical Engineer, Tritium Removal & Heavy Water Management
September 2018 – December 2019

- Developed a C++/VBA simulation to forecast nuclear reactor radioactivity accumulation to balance decontamination schedules for Canada's fleet of nuclear reactors as a strategical initiative for Ontario Power Generation
- Prepared procedures for technicians to analyze aging tritium hydride storage beds for the quantification of Helium-3 inventories. Discovered high purities of Helium-3 for recovery by utilizing a high vacuum system & residual gas analyzer. Accountable for the discovery & quantification of ∼$5M worth of Helium-3 in market value

Personal Projects

10-Q/10-K Alphas : Created a scraper to filter all 10-Q/K earning reports to predict future returns by algorithmic scoring of fraudulent statements from SEC EDGAR. Deployed algorithm with an AUM of $3 M USD. Used python, pandas, FIX, XBRL

McKinsey & Company Hackathon : Analyzed if a requested loan is approved or not, based on the features of the applicant (e.g., income, gender, city), and that of the loan (e.g., loan amounts, interest rates). Used XGBoost / gradient boosting for analysis

Scotiabank Hackathon : Concepted a mobile game to help reduce user debt with in-app purchases routed back to saving accounts. Participated with seven software engineers and graphics designers within 48 hours to win a $10,000 prize

Education

University of Ontario Institute of Technology *2018*
Bachelor of Engineering – Nuclear Engineering

Research : Created a MATLAB application to track AR device user location in GPS denied environments using MEMS input sensor data. Developed an autonomous radiation detecting roomba to decontaminate areas.